Studypalooza Problems | CHEM 1211K

Compounds and the Mole I

3.A.1. In this series of problems we will examine a few structures on the Crystallography Open Database (COD).

A. Visit this link and examine the three-dimensional structure that appears. Hover your mouse over each atom to see its chemical symbol. This structure contains both ionic and covalent bonds. List the different types of covalent bonds in the structure (e.g., C–C). Then, find the lone cation in the structure and determine its charge.

   C—C, C—H, C—Si, C—O, C—N, Mg^{2+}

B. Visit this link and examine the three-dimensional structure that appears. Examine the AlF_5 molecules. What is the net charge on each molecule? How do you know? What is the net charge on the $HN(CH_2CH_2NH_3)_3$ molecule?

   The net charge on $HN(CH_2CH_2NH_3)_3$ is +4 (note the four tetracoordinate nitrogen atoms), so the charge on each AlF_5 unit must be –2.

C. Visit this link and examine the three-dimensional structure that appears. The title of the paper reporting this structure mentions a "heavy alkaline earth" metal. What is the identity of this metal and what is its charge in the structure? What other charged atoms are in the structure?

   As an alkaline earth metal, the charge must be +2. The atomic symbol "Sr" in the structure confirms that the element is strontium.

3.A.2. Classify each of the following acids as a binary acid or an oxyacid and determine its name. As you generate the name, document your thought process in as much detail as you can.

A.	HI	binary acid	hydroiodic acid (hydrogen iodide)
B.	$HBrO_4$	oxyacid	perbromic acid
C.	HBr	binary acid	hydrobromic acid (hydrogen bromide)
D.	H_2S	binary acid	hydrosulfuric acid (hydrogen sulfide)
E.	H_3PO_3	oxyacid	phosphorous acid

3.A.3. Each of the following compounds can be used as a source of phosphorus in the synthesis of H_3PO_3. Determine whether each is an ionic or covalent compound and name them. Separate the ionic compounds into their component ions.

A. K_2HPO_3 ionic potassium hydrogen phosphite

B. PCl_3 covalent phosphorus trichloride

C. P_4O_6 covalent tetraphosphorus hexoxide

3.A.4. The anion BF_4^- has acts in many ways like an oversized halide anion. The conjugate acid of this anion, HBF_4, is of interest as a strong acid (like HCl, etc.).

A. The name of BF_4^- is “tetrafluoroborate.” From this information, generate a name for the acid HBF_4.

Tetrafluoroboric acid. The suffix “-ate” in the anion becomes “-ic acid.”

B. Pure HBF_4 decomposes to form HF and BF_3. Provide chemical names for each of the products of this reaction.

hydrofluoric acid (hydrogen fluoride) and boron trifluoride

Compounds and the Mole II

3.B.1. The fact that a mole corresponds to 6.022×10^{23} (Avogadro’s number, N_A) objects is well known. But where does this number come from? Through the following series of problems, we’ll come to understand the mole better as a concept.

A. One carbon-12 atom has a mass of 12.00 atomic mass units (12.00 u). Using Wolfram Alpha, calculate the mass of Avogadro’s number of carbon-12 atoms. (Wolfram Alpha is an intelligent calculator. You can type in “Avogadro’s number” and it will understand what you mean quantitatively.)

12.00 grams—this follows from the definition of the atomic mass unit!

B. Repeat the calculation for Avogadro’s number of oxygen-16 atoms (15.99 u) and bromine-79 atoms (78.92 u). Do you notice a pattern in the masses of N_A atoms?

Avogadro’s number of oxygen-16 atoms weighs 15.99 grams and the same number of bromine-79 atoms weighs 78.92 grams.

C. The pattern you’re seeing is general for atoms and molecules. State the pattern as a rule that relates mass on the atomic or molecular scale to mass on the scale of 1 mole.

From the definition of the mole, it follows that 1 mole of a substane has a mass in grams equal to the mass of a single formula unit in atomic mass units.

3.B.2. We have distinguished between the terms *formula mass* and *molecular mass*. While not profoundly different in practice, these terms refer to compounds that look very different on the sub-microscopic scale. For each compound below, indicate whether the term “formula mass” or “molecular mass” is more appropriate for describing the mass of the formula unit. Then, calculate that mass in atomic mass units (u).

A. $CaTiO_3$ formula mass 135.94 u

B. $C_{21}H_{23}NO_5$ molecular mass 369.41 u

C. Al_2O_3 formula mass 101.96 u

D. $Be_3Al_2(SiO_3)_6$ formula mass 537.50 u

3.B.3. Desmopressin is a drug that promotes the re-uptake of water by cells, raising blood pressure and decreasing the frequency of urination.

A. Desmopressin contains 51.67% C, 6.03% H, 18.34% N, 17.96% O, and 6.00% S by mass. Determine the empirical formula of desmopressin from this data.

$C_{23}H_{32}N_7O_6S$

B. An independent experiment confirmed that a sample of 1.0692 g of desmopressin contains 1.000 mmol of desmopressin molecules. What is the molar mass of desmopressin? What is its molecular formula?

$C_{46}H_{64}N_{14}O_{12}S_2$. Note that “mmol” refers to *milli*moles!

3.B.4. The combustion of an unknown hydrocarbon (a compound containing *only* carbon and hydrogen) produced 8.80 g of CO_2 and 4.50 g of H_2O. The molar mass of the compound, determined in a separate experiment, is 58.124 g/mol. What is the molecular formula of the compound?

The empirical formula implied by this data is C_2H_5, the molar mass of which is about 29 g/mol. Because the actual molar mass is twice this value, the molecular formula is C_4H_{10}.

Chemical Reactions and Aqueous Solutions I

4.A.1. Classify each of the reactions below using the reaction classes identified in this unit.

A. $Si(s) + 2\ Cl_2(g) \rightarrow SiCl_4(l)$ synthesis

B. $2\ C_4H_{10}(g) + 13\ O_2(g) \rightarrow 8\ CO_2(g) + 10\ H_2O(g)$ combustion

C. $Pb(NO_3)_2(aq) + 2\ KI(aq) \rightarrow PbI_2(s) + 2\ KNO_3(aq)$ double replacement

4.A.2. What is the driving force for reaction 1C?

Formation of a solid: precipitation of the insoluble ionic compound lead(II) iodide drives the reaction.

4.A.3. To better understand and predict how and why reactions happen, we often think about what's going on at the molecular or sub-microscopic level. For each of the reactions below, draw molecular-scale pictures of the reactants and products. Make an effort to balance the elements in both pictures.

A. $2\ H_2(g) + O_2(g) \rightarrow 2\ H_2O(l)$

B. $MgI_2(aq) + 2\ AgNO_3(aq) \rightarrow 2\ AgI(s) + Mg(NO_3)_2(aq)$

C. $CaCO_3(s) \rightarrow CaO(s) + CO_2(g)$

Try this one on your own!

4.A.4. When aqueous sodium carbonate (Na_2CO_3) is mixed with an aqueous solution of acetic acid ($HO_2C_2H_3$), a reaction occurs that produces carbon dioxide (CO_2), water (H_2O), and sodium acetate ($NaO_2C_2H_3$).

This reaction occurs in two stages: an initial formation of carbonic acid (H_2CO_3) followed by conversion of carbonic acid to water and carbon dioxide. *Unbalanced* equations for these processes are below.

_1_ $Na_2CO_3(aq)$ + _2_ $HO_2C_2H_3(aq) \rightarrow$ _2_ $NaO_2C_2H_3(aq)$ + _1_ $H_2CO_3(aq)$

_1_ $H_2CO_3(aq) \rightarrow$ _1_ $H_2O(l)$ + _1_ $CO_2(g)$

Provide coefficients in both chemical equations so that the overall conversion of sodium carbonate to sodium acetate, water, and carbon dioxide is balanced.

A. How would you classify the first reaction? What is its driving force?

   This is an example of a double-replacement reaction, driven by formation of a more stable anion on the product side (note that this is beyond the scope of CHEM 1211K).

B. How would you classify the second reaction?

   This is a decomposition reaction.

4.A.5. Melanoidins are complex structures formed when wet barley malt is heated in a kiln during the beermaking process. To model the formation of melanoidins, an α-hydroxy acetaldehyde ($C_2H_4O_2$) was combined with methylamine (CH_5N). The products were water (H_2O) and an organic compound with the formula $C_6H_{10}N_2$. Balance this chemical equation.

_2_ $C_2H_4O_2$ + _2_ $CH_5N \longrightarrow$ _1_ $C_6H_{10}N_2$ + _4_ H_2O

Chemical Reactions and Aqueous Solutions II

4.B.1. In this unit we highlighted the difference between *dissolution* and *dissociation*.

A. Provide an example of a compound that does not undergo dissociation when dissolved in water. Write a balanced chemical equation with state indicators for the dissolution of this compound in water, looking up formulas where necessary.

   Ethanol! $C_2H_5OH(l) \rightarrow C_2H_5OH(aq)$. Any other molecular compound that is not an acid will also work here.

B. Provide an example of a compound that *does* undergo dissociation when dissolved in water. Write separate balanced chemical equations for the dissolution and dissociation processes.

   Potassium hydroxide. $KOH(s) \rightarrow KOH(aq)$ is the dissolution process
   and $KOH(aq) \rightarrow K^+(aq) + OH^-(aq)$ is the dissociation process.

C. For your chosen compound in B, draw molecular-level pictures of the solution immediately following dissolution and following dissociation.

Try this one on your own!

4.B.2. Determine whether each of the following compounds forms a strong or weak electrolyte when dissolved in water. Provide a justification in each case.

A. NaOH strong electrolyte

B. $NiBr_2$ strong electrolyte

C. H_2SO_4 strong electrolyte

D. $C_6H_{12}O_6$ (sucrose) non-electrolyte

E. $HO_2C_2H_3$ (acetic acid) weak electrolyte

F. HCl strong electrolyte

Chemical Reactions and Aqueous Solutions III

4.C.1. The solubility rules are qualitative guidelines that enable us to predict whether a given ionic solid is soluble in water or not.

A. Use the solubility rules to construct six ionic compounds involving different ions that are soluble in water.

There are tons of possibilities here, but here are six examples: $LiNO_3$, NaOH, KBr, NH_4Cl, SrI_2, K_3PO_4.

B. Use the solubility rules to construct six ionic compounds involving different ions that are insoluble in water.

There are tons of possibilities here, but here are six examples: $Co(OH)_2$, $CaCO_3$, AgCl, $BaCrO_4$, MgF_2, Ag_3PO_4.

4.C.2. Lead(II) nitrate reacts with sodium chloride in aqueous solution to form a precipitate.

A. Write balanced molecular, ionic, and net ionic equations for this process with state indicators.

Molecular. $Pb(NO_3)_2(aq) + 2\ NaCl(aq) \rightarrow PbCl_2(s) + 2\ NaNO_3(aq)$

Ionic. $Pb^{2+}(aq) + 2\ NO_3^-(aq) + 2\ Na^+(aq) + 2\ Cl^-(aq) \rightarrow PbCl_2(s) + 2\ Na^+(aq) + 2\ NO_3^-(aq)$

Net Ionic. $Pb^{2+}(aq) + 2\ Cl^{-}(aq) \rightarrow PbCl_2(s)$

B. If you haven't already, identify the spectator ions in the reaction.

The spectator ions are Na^{+} and NO_3^{-}.

4.C.3. For each combination of aqueous solutions of ionic salts below, predict the products and write a balanced molecular equation. If no reaction occurs, just write "no reaction."

A. $LiNO_3(aq) + NaOH(aq) \rightarrow$ no reaction

B. $Na_2S(aq) + FeCl_2(aq) \rightarrow$ $FeS(s) + 2\ NaCl(aq)$

C. $3\ CaBr_2(aq) + 2\ (NH_4)_3PO_4(aq) \rightarrow$ $Ca_3(PO_4)_2(s) + 6\ NH_4Br(aq)$

D. $Co(NO_3)_2(aq) + 2\ KOH(aq) \rightarrow$ $Co(OH)_2(s) + 2\ KNO_3(aq)$

Stoichiometry I

5.A.1. In several coming problems, we'll explore the process of creating a desmopressin solution outlined in this video. We previously encountered the anti-diuretic drug desmopressin ($C_{46}H_{64}N_{14}O_{12}S_2$, 1069.2 g/mol) above.

This compound is commonly treated with acetic acid ($HC_2H_3O_2$, 60.1 g/mol) to produce desmopressin acetate ($C_{48}H_{68}N_{14}O_{14}S_2$, 1129.3 g/mol). Desmopressin and acetic acid react in a 1:1 molar ratio.

A. How many moles of desmopressin are present in a 120. mg sample?

0.000112 mol or 0.112 mmol

B. What mass of acetic acid is necessary to completely consume this amount of desmopressin?

Because desmopressin and acetic acid react in a 1:1 ratio, 0.000112 mol of acetic acid are required. The mass of this amount of acetic acid is 0.00675 g or 6.75 mg.

5.A.2. The compound α,α'-dibromo-*o*-xylene ($C_8H_8Br_2$, 264.0 g/mol) reacts with lithium metal to give a product containing 16 carbons as well as lithium bromide. A balanced chemical equation for this process is given below.

$2\ C_8H_8Br_2 + 4\ Li \rightarrow C_{16}H_{16} + 4\ LiBr$

In one synthesis of $C_{16}H_{16}$, 101 g of $C_8H_8Br_2$ were combined with 6.63 g of lithium metal.

A. Determine the number of moles of each reactant.

0.383 mol $C_8H_8Br_2$ and 0.960 mol Li

B. Determine the limiting reactant.

$C_8H_8Br_2$ is limiting because the amount of preparable $C_{16}H_{16}$ product from the lithium (0.960 mol / 4) is greater than the amount of preparable $C_{16}H_{16}$ product from the $C_8H_8Br_2$ (0.383 mol / 2).

5.A.3. Barium nitride (Ba_3N_2) and water react to form barium hydroxide ($Ba(OH)_2$) and ammonia (NH_3).

A. Write a balanced chemical equation for this process.

$Ba_3N_2(s) + 6\ H_2O(l) \rightarrow 3\ Ba(OH)_2(s) + 2\ NH_3(aq)$

B. What is the maximum mass of barium hydroxide that can be produced from a mixture of 50.0 g of barium nitride and 20.0 g of water?

58.4 g; see this solution. Pretty impressed with ChatGPT, to be honest!

C. Under the conditions of the previous problem, assuming the reaction goes to completion, what mass of the excess reactant is left behind?

7.72 g; see this solution.

5.A.4. Calcium hydroxide ($Ca(OH)_2$) is formed from the reaction of calcium oxide (CaO) with water (H_2O).

A. Write a balanced chemical equation for this process.

$CaO(s) + H_2O(l) \rightarrow Ca(OH)_2(s)$

B. What mass of calcium hydroxide can be produced from a mixture of 25.0 g of calcium oxide and 12.0 g of water?

33.1 g; see this solution.

C. Under the conditions of the previous problem, assuming the reaction goes to completion, what mass of the excess reactant is left behind?

3.96 g; see this solution.

Stoichiometry II

5.B.1. The diphosphorus pentoxide used to produce phosphoric acid for cola drinks is prepared by burning phosphorus in oxygen.

$P_4 + 5\ O_2 \longrightarrow P_4O_{10}$

A. What is the limiting reactant when 0.200 mol P_4 are combined with 0.200 mol O_2?

O_2; see this solution.

B. What is the percent yield of diphosphorus pentoxide if 10.0 grams of P_4O_{10} are obtained at the end of the reaction?

88%; see this solution.

5.B.2. In the last set of problems we considered the reaction below.

$2\ C_8H_8Br_2 + 4\ Li \rightarrow C_{16}H_{16} + 4\ LiBr$

In one synthesis of $C_{16}H_{16}$, 101 g of $C_8H_8Br_2$ were combined with 6.63 g of lithium metal. The actual yield of $C_{16}H_{16}$ was 19.6 g. What is the percent yield of $C_{16}H_{16}$?

49.2%; see this solution. Note that ChatGPT screws up in the first attempt; read the full solution carefully!

5.B.3. Brewing water is commonly "treated" with one or more ionic salts to provide nutrients for yeast, adjust pH, or improve flavor. In one treatment, 0.058 moles of calcium in the form of calcium chloride ($CaCl_2$) are dissolved in 4 gallons of water.

A. Write a balanced chemical equation for the dissolution of calcium chloride in water.

$CaCl_2(s) \rightarrow Ca^{2+}(aq) + 2\ Cl^-(aq)$

B. How many moles of chloride ions are added to the water in the process? What are the molarities of calcium and chloride ions?

0.116 mol Cl^-. Note the 1:2 molar ratio of $CaCl_2$ to Cl^-.

$[Cl^-] = 0.00766$ M; $[Ca^{2+}] = 0.00383$ M. One gallon is basically four liters (you should convert more precisely, of course).

C. What mass of calcium chloride is needed to create this solution?

6.44 g.

5.B.4. Let's continue to explore the process of creating a desmopressin solution outlined in this video. We previously encountered the anti-diuretic drug desmopressin ($C_{46}H_{64}N_{14}O_{12}S_2$, 1069.2 g/mol) in Chapter 3. If you haven't already, watch the complete video before tackling the problems below.

A. A mass of 120. mg desmopressin acetate ($C_{48}H_{68}N_{14}O_{14}S_2$, 1129.3 g/mol) is first dissolved in 100. mL of water. Assuming the final solution volume is 100. mL, what is the molarity of desmopressin in this solution?

0.00106 M.

B. A volume of 1.0 mL of this solution is added to "vehicle," a mixture of simple and cherry syrup, to create a final solution with a volume of 60.0 mL. What is the molarity of desmopressin in the resulting solution?

1.77×10^{-5} M.

C. Verify that this molarity corresponds to 0.1 mg of desmopressin per 5 mL of solution as reported on the video.

A mass of 0.1 mg desmopressin in 5 mL of solution would amount to 20 mg desmopressin in 1 L of solution; 20 mg desmopressin does correspond to 1.77×10^{-5} mol.

Stoichiometry III

5.C.1. Calculate each of the following quantities for additional practice with solutions.

A. Grams of solute in 175.8 mL of 0.0565 M calcium acetate

1.48 g.

B. Molarity of a 500. mL solution containing 3.38 g potassium iodide

0.0407 M or 40.7 mM.

C. Moles of solute in 3.011 L of 0.850 M sodium cyanide

2.56 mol.

5.C.2. Arrange the solutions in order of increasing number of particles in solution assuming that the volumes of the solutions are the same: 0.50 M Na_3PO_4, 1.0 M C_6H_{14}, 1.0 M KCl, 1.0 M $Ca(NO_3)_2$.

least 1.0 M C_6H_{14}, 1.0 M KCl = 0.50 M Na_3PO_4, 1.0 M $Ca(NO_3)_2$ **most**

5.C.3. The density of ethanol (C_2H_6O) is 0.789 g/mL. Ethanol reacts with acetic acid ($HC_2H_3O_2$) in a 1:1 molar ratio to yield ethyl acetate ($C_4H_8O_2$) and water (H_2O). What volume of a 2.0 M acetic acid solution is required to completely consume 500. mL of ethanol?

4.28 L; see this solution.

5.C.4. The weak acid hydrazoic acid (HN_3) can be titrated by sodium hydroxide (NaOH) to determine its concentration in solution. The two react according to the balanced chemical equation below.

$HN_3 + NaOH \rightarrow NaN_3 + H_2O$

A. What volume of 0.20 M NaOH is required to completely consume the hydrazoic acid in 25.0 mL of a 0.50 M HN_3 solution?

   62.5 mL.

B. After the two solutions have been combined and the reaction has run its course, what is the molarity of NaN_3 in the resulting solution? Assume that the reaction goes to completion.

   0.143 M.

C. What is $[HN_3]$ in a 10.0-mL sample that requires 30.0 mL of 0.15 M NaOH to completely consume the HN_3?

   0.45 M.

5.C.5. In a common medical laboratory determination of the concentration of free chloride ion in blood serum, a serum sample is titrated with a $Hg(NO_3)_2$ solution.

$2\ Cl^- + Hg(NO_3)_2 \longrightarrow 2\ NO_3^- + HgCl_2$

What is $[Cl^-]$ in a 0.25-mL sample of serum that requires 1.46 mL of 8.25×10^{-4} M $Hg(NO_3)_2$ to fully consume the Cl^- in the sample?

0.00964 M or 9.64 mM.

Thermochemistry I

6.A.1. In aqueous solution, lithium sulfate (Li_2SO_4) reacts with barium chloride ($BaCl_2$) to form solid barium sulfate ($BaSO_4$) and aqueous lithium chloride (LiCl). Draw a diagram of this situation and label the system and surroundings, keeping in mind that, as chemists, we are interested in the energy change as the reactants become products.

$Li_2SO_4(aq) + BaCl_2(aq) \rightarrow BaSO_4(s) + 2\ LiCl(aq)$

6.A.2. Provide definitions and examples of the following types of systems. Tons of options here; try this on your own!

A. An *isolated* system

B. A *closed* system that can exchange *only heat* with its surroundings

C. A *closed* system that can exchange *heat or work* with its surroundings

D. An *open* system

6.A.3. Inside a refrigerator, a refrigerant gas absorbs 80 cal of heat from the refrigeration compartment and its internal energy increases by 450 J.

A. Draw a diagram of this situation that includes the refrigerant gas line and the refrigeration compartment. Label the system and surroundings, keeping in mind that, as chemists, we are interested in the internal energy of the gas.

The most important thing here is to think of the refrigerant gas particles *only* as the system. Everything else is surroundings, including the walls of the gas line and compartment.

B. Calculate the work done on the gas during this process in Joules, considering the first law of thermodynamics.

Just an exercise in unit conversion and application of the first law. $w = \Delta U - q = 115.3$ J.

6.A.4. Consider two identical cans of beans **A** and **B** starting at 20 °C. The cans are able to exchange energy with their surroundings but no particles can enter or leave the cans. A timer is started and can **A** is cooled to 5 °C and then heated to 80 °C. Simultaneously, can **B** is immediately heated to 80 °C. The timer is stopped when can **A** reaches 80 °C, at which point can **B** is already at this temperature.

A. Consider the processes for **A** and **B** that started when the timer was started and ended when the timer was stopped. List two examples of quantities that are *equal* for both processes and explain why they are equal. (*Hint:* the values depend only on the initial and final conditions and not on the process leading from one to the other.)

Any change in a state function will work here, such as ΔT, ΔH, ΔU, etc.

B. Consider the processes for **A** and **B** that started when the timer was started and ended when the timer was stopped. List two examples of quantities that are *unequal* for both processes and explain why they are unequal.

Any path function will work here, such as the time, total heat (the heat capacities of the beans will vary slightly with temperature!), or work (if the cans can expand or contract).

Thermochemistry II

6.B.1. Inside a refrigerator, an ideal gas expands through a valve at a constant pressure of 40 psi from 5.0 mL to 500.0 mL. At the same time, the gas absorbs 125 J of heat from its surroundings.

A. Draw a diagram of this situation that includes the gas line, the valve, and the surroundings (the refrigeration compartment). Label the system and surroundings, keeping in mind that, as chemists, we are interested in the internal energy of the gas.

See problem 6.A.3. As above, key is to think of *only* the gas particles as the system of interest.

B. Label the directions of heat and work flows. What are the signs of heat and work from the perspective of the system?

Heat is positive (heat is absorbed by the gas) but work is negative (the gas expands).

C. Calculate the heat and work and use them to calculate the internal energy change of the gas during this process in Joules.

From $w = -P\,\Delta V$, the work is –136.5 J. Using $\Delta U = q + w$, the change in internal energy is –11.5 J.

6.B.2. Imagine that the (tiny) refrigeration compartment in the last problem was filled with 0.010 L of water at 20.0 °C. What will the temperature of the water be when the process described in problem 1 is complete?

17.0 °C. Use the 125 J of heat absorbed by the gas and assume all this comes from the water.

6.B.3. A sample of coal is burned in an apparatus with a calorimeter constant of 1.3 kJ/°C. When a sample weighing 0.367 g is used, the temperature change is +8.75 °C. What is the energy density of this coal in units of kJ/g?

31.0 kJ/g.

6.B.4. A 61.0 g sample of hot metal, which initially has a temperature of 120.0 °C, is plunged into 100. g of water that is initially at 20.0 °C. The metal cools down and the water heats up until they reach a common temperature of 26.39 °C.

A. Draw a diagram of this situation, label the system and surroundings, and label the direction(s) of relevant heat flow(s).

B. Before doing any calculations, is the specific heat of the metal less than or greater than the specific heat of water?

$c(H_2O) > c(\text{metal})$.

C. Calculate the specific heat of the metal, using 4.184 J/g·°C as the specific heat of the water.

0.468 J/g·°C.

6.B.5. Predict whether each of the following processes is endothermic or exothermic. Consider the definition of enthalpy (*H*).

A. The temperature of an ideal gas increases inside a closed, rigid container. endothermic

B. An ideal gas is compressed at constant pressure and constant internal energy. exothermic

C. The temperature of an ideal gas decreases inside a closed, rigid container. exothermic

D. An ideal gas expands at constant pressure and constant internal energy. endothermic

Thermochemistry III

6.C.1. When 4 moles of iron metal react with 3 moles of oxygen gas to form 2 moles of iron(III) oxide, the reaction releases 1652 kJ of heat at constant pressure.

A. Write a balanced thermochemical equation for this process.

$4\ Fe(s) + 3\ O_2(g) \rightarrow 2\ Fe_2O_3(g) + 1652\ kJ$

B. How much heat is released when 10.0 g of iron metal and 3.00 g of oxygen gas react? Assume that the reaction goes to completion.

51.7 kJ. Be sure to find the limiting reactant first and use that as the basis for this calculation!

6.C.2. When solutions contain silver ions and chloride ions are mixed, silver chloride precipitates. One mole of reaction events releases 65.5 kJ of heat at constant pressure.

$Ag^+(aq) + Cl^-(aq) \rightarrow AgCl(s)$

A. Write a balanced thermochemical equation for this process.

$Ag^+(aq) + Cl^-(aq) \rightarrow AgCl(s) + 65.5\ kJ$

B. Determine the change in enthalpy when 9.25×10^{-4} moles of AgCl *solid* dissolves in one liter of water.

The process described here is the *reverse* of the reaction in part A! +60.6 J.

6.C.3. Given the following data, calculate ΔH°_{rxn} for the target reaction below.

Target: $H_2(g) + Br_2(l) \rightarrow 2\ HBr(g)$

$NH_3(g) + HBr(g) \rightarrow NH_4Br(s)$	$\Delta H^{\circ}_{rxn} = -188.32$ kJ
$N_2(g) + 3\ H_2(g) \rightarrow 2\ NH_3(g)$	$\Delta H^{\circ}_{rxn} = -92.22$ kJ
$N_2(g) + 4\ H_2(g) + Br_2(l) \rightarrow 2\ NH_4Br(s)$	$\Delta H^{\circ}_{rxn} = -541.66$ kJ

–72.8 kJ.

6.C.4. Inside an internal combustion engine, gaseous *n*-octane (C_8H_{18}) reacts with oxygen gas (O_2) to form carbon dioxide gas (CO_2) and water vapor (H_2O).

A. Write a balanced chemical equation for this process.

$2\ C_8H_{18}(g) + 25\ O_2(g) \rightarrow 16\ CO_2(g) + 18\ H_2O(g)$

B. Use the NIST Chemistry WebBook to look up the enthalpies of formation of gaseous *n*-octane, gaseous carbon dioxide, and water vapor. Explain why looking up the enthalpy of formation of gaseous oxygen is unnecessary.

Gaseous oxygen is an element in its natural state, so its enthalpy of formation is zero.

C. Use the enthalpies of formation you looked up and the balanced equation you wrote to calculate ΔH°_{rxn} for the combustion of gaseous *n*-octane.

–10230 kJ. This reaction is *extremely* exothermic!

Gases I

7.A.1. An understanding of gas laws is critical in the carbonation process. To "force-carbonate" beer, the liquid is placed in a sealed vessel (keg) and the headspace is filled with carbon dioxide at high pressure. This may be done at room temperature or after cooling the beer to refrigerator temperatures.

A. Imagine that 5 gallons of beer are force-carbonated using 50 psi of carbon dioxide at room temperature (20.0 °C). The keg is then placed in a refrigerator ("kegerator") and cooled to 3.0 °C. Assuming no change in the solubility of carbon dioxide in the liquid and no changes in volume or number of moles, what will the pressure of carbon dioxide be at the cold temperature?

   47.1 psi. Use the relation between pressure and temperature.

B. After cooling, the pressure of carbon dioxide in a keg is *much* smaller than the value in part A because the solubility of a gas in water increases as the temperature drops. As the temperature falls, more carbon dioxide dissolves in the beer.

   Imagine the headspace above the beer has a volume of 1 gallon (3.79 L). If the measured pressure at 3 °C is 12.1 psi, what amount in moles of additional carbon dioxide has dissolved in the beer upon cooling? You'll need to use the theoretical pressure from the last problem. The ideal gas constant is 0.319 psi·gallons/K·mol.

   0.398 mol CO_2.

7.A.2. A gaseous mixture made from 10.0 g of oxygen and 5.0 g of methane is placed in a 10.0 L vessel at 25 °C. What is the partial pressure of each gas (in atm), and what is the total pressure in the vessel?

0.764 atm O_2 and 0.764 atm CH_4, with total pressure 1.528 atm (the sum of the two partial pressures, in accordance with Dalton's law).

7.A.3. A mixture of 35.6 L of $NH_3(g)$ and 40.5 L of $O_2(g)$ are mixed at standard temperature and pressure (STP) and ignited. The reaction produces gaseous water and NO(*g*). After the products return to STP, how many grams of NO are present?

43.5 g NO. Write a balanced equation and find the limiting reactant to get started.

7.A.4. Check out the simulation of an ideal gas here: http://falstad.com/gas/. The distribution of particle speeds is shown near the bottom of the simulation. Use the dropdown menu to explore different initial configurations of the gas molecules. Make sure to try out the "Two Gases" items. Determine whether each statement below is true or false.

A. The distribution of speeds at long simulation times depends profoundly on the initial configuration of gas molecules. false
B. Heating the gas causes the distribution of speeds to "tighten up" or become "spikier." false

C. Once the distribution of speeds converges to a Maxwell distribution, fluctuations above or below the theoretical distribution are not observed. false
D. When two gases with different masses are mixed randomly in the same container, they converge to different speed distributions. true
E. When two gases with different masses are mixed randomly in the same container, they converge to different energy distributions. false

Gases II

7.B.1. Arrange the following samples of gas in order of increasing root mean square speed and explain your ordering.

A. O_2 at 700 K
B. O_2 at 400 K
C. H_2 at 700 K
D. H_2 at 400 K

slowest B, D, A, C **fastest**

Speed depends on temperature (hotter gases contain faster-moving particles) and particle mass (heavier particles move more slowly).

7.B.2. Arrange the following samples of gas in order of increasing kinetic energy and explain your ordering.

A. O_2 at 700 K
B. O_2 at 400 K
C. H_2 at 700 K
D. H_2 at 400 K

lowest KE B = D, A = C **highest KE**

Kinetic energy is *only* dependent on temperature. The kinetic energies of O_2 and H_2 at a common temperature are equal!

7.B.3. Whoops—your toddler poked a tiny hole in your keg. Carbon dioxide, nitrogen, and oxygen gases in the headspace immediately started effusing through the hole. Which of the three gases effuses at the highest rate?

Nitrogen effuses most rapidly because it is the lightest of the three gases.

7.B.4. If it takes 67 minutes for a given volume of H_2 to effuse, how long will it take an equal volume of UF_6 to effuse under the same conditions of temperature and pressure?

889 min. Apply Graham's law of effusion.

7.B.5. If it takes 67 minutes for a given volume of H_2 to effuse at 25 °C, how long will it take the same volume to effuse if the temperature is increase to 37 °C? **NOTE:** the rate of effusion is inversely proportional to $T^{1/2}$, where T is in Kelvin.

65.7 min.

7.B.6. An unknown gas effuses at a rate that is 1.5 times faster than chlorine gas (at the same temperature). What is the molar mass of the unknown gas?

31.5 g/mol.

7.B.7. What evidence could you provide to indicate that, unlike ideal gases, real gas particles do have some attractions to each other?

Think through this one on your own!

Quantum Model of the Atom I

8.A.1. The Voyager 1 spacecraft was launched in 1977 to explore the outer Solar System. In December 2012, it was 1.85×10^{13} m from Earth and still sending and receiving data via radio signals. At this distance, how long did it take signals from Voyager 1 to reach Earth?

17.1 hours.

8.A.2. The frequencies of light associated with traffic signals are listed below. For each, calculate the wavelength in nanometers, the energy of a *single* photon in Joules, and the energy of a *mole* of photons in kilojoules per mole. Finally, list the colors associated with each frequency.

5.75×10^{14} Hz	5.15×10^{14} Hz	4.37×10^{14} Hz
521 nm (green)	582 nm (yellow)	686 nm (red)
3.81×10^{-19} J	3.41×10^{-19} J	2.90×10^{-19} J
230 kJ/mol	206 kJ/mol	174 kJ/mol

8.A.3. A laser that emits light energy in pulses of short duration has a frequency of 4.69×10^{14} s^{-1} and deposits 1.3×10^{-2} J of energy during each pulse. How many photons does each pulse deposit?

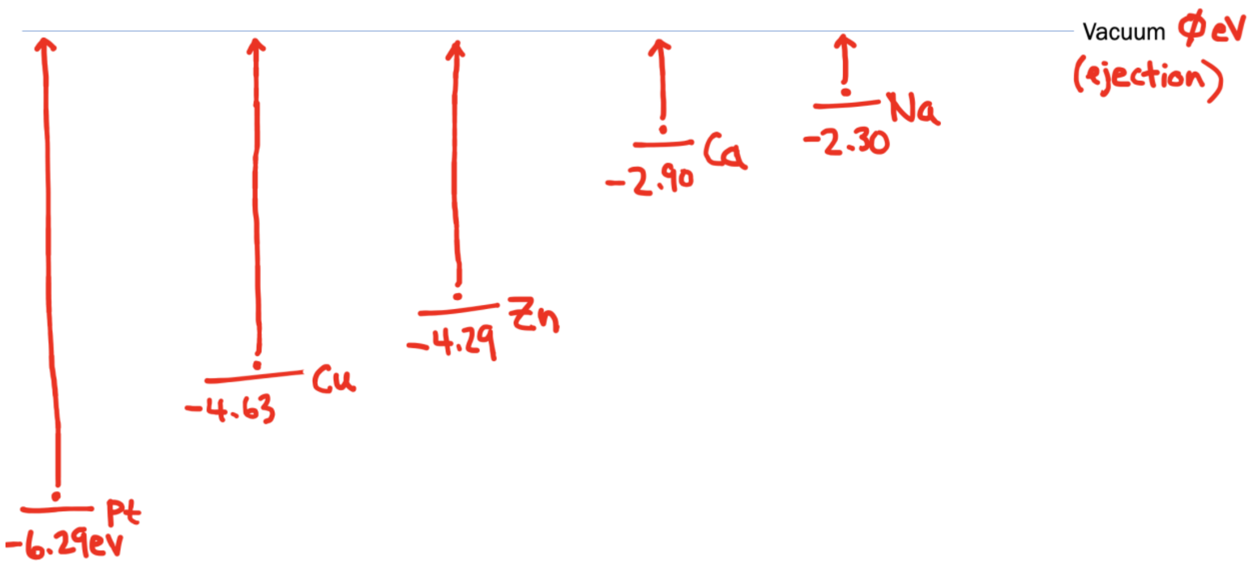
4.18×10^{16} photons.

8.A.4. The photoelectric effect refers to the ejection of electrons from a metal by light impinging on the metal. There is a minimum frequency of light needed to cause ejection of an electron; this frequency is unique to each metal and related to an energy called the **work function (ϕ)**.

A. Using [this simulation], choose a metal and measure its work function. To do this, determine the *maximum* wavelength that causes ejection of an electron and calculate the energy of the light at this wavelength in electron-Volts (eV).
   hc = 1240 eV·nm.

   Na: 540 nm, 2.30 eV
   Zn: 289 nm, 4.29 eV
   Cu: 268 nm, 4.63 eV
   Pt: 197 nm, 6.29 eV
   Ca: 428 nm, 2.90 eV

B. Using the results of your peers along with your own, prepare an energy diagram that shows the relative energies of the highest-energy electrons in each metal.



Quantum Model of the Atom II

8.B.1. The Bohr model of the atom was an important precursor to the fully quantum model that we use today. Bohr based his model on the observation that atoms only absorb and emit particular wavelengths of light. Draw a picture of the Bohr model of the hydrogen atom and use arrows to show how the electron changes in energy and position when the atom absorbs or emits a photon.

8.B.2. The Rydberg equation was developed as an empirical model that explains the wavelengths λ of light emitted by excited hydrogen atoms.

$$\frac{1}{\lambda} = R_{\mathrm{H}}\left(\frac{1}{n_f^2} - \frac{1}{n_i^2}\right)$$

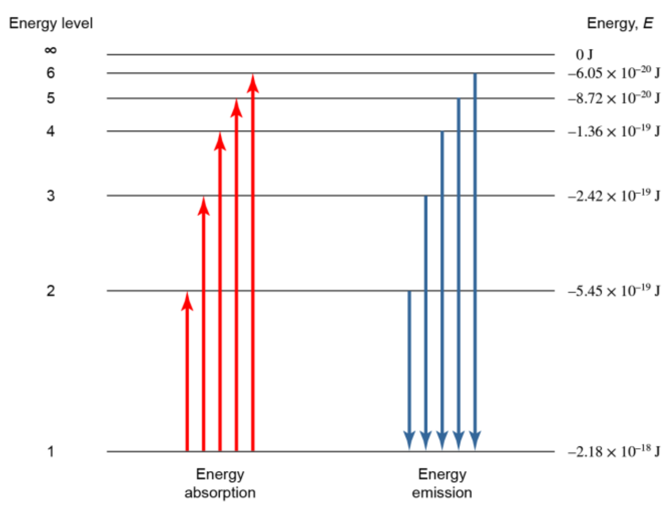
A. Label each of the variables on the right-hand side of this equation and provide a value for R_{H} in appropriate units.

   Try this one on your own! R_H (we've also called it R_∞) here has a value of 1.097×10^7 m^{-1}.

B. The Rydberg equation can be written in an equivalent form with the energy of the emitted photon on the left-hand side instead of its wavelength. Show how this can be done starting from the equation above and use your approach to calculate a value for R_{H} in units of electron-Volts (eV).

   R_H (we've also called it R_∞) has a value of 13.6 eV here. It's worth noting that this is the ionization energy of hydrogen (see ptable.com) and this is *NOT* a coincidence!

C. On the diagram below, label the ground state and excited states. Order the emissive transitions from longest to shortest wavelength of light emitted.



$n = 1$ is the ground state and all states above it ($n = 2, 3, 4, \ldots$) are excited states. The $2 \rightarrow 1$ transition emits light of the *longest* wavelength.

8.B.3. Atomic electrons are arranged in shells indexed by a positive integer *n* that we will call the *principal quantum number*. Determine the number of electrons that each shell can hold by completing the table below. Start with the second column from the left and fill out the table left to right.

Shell (*n*)	Subshells	Orbitals within subshell	Total number of electrons in subshell	Total number of electrons in shell
1	1*s*	1	2	2
2	2*s*	1	2	8
	2*p*	3	6	
3	3*s*	1	2	18
	3*p*	3	6	
	3*d*	5	10	
4	4*s*	1	2	32
	4*p*	3	6	
	4*d*	5	10	
	4*f*	7	14	

Quantum Model of the Atom III

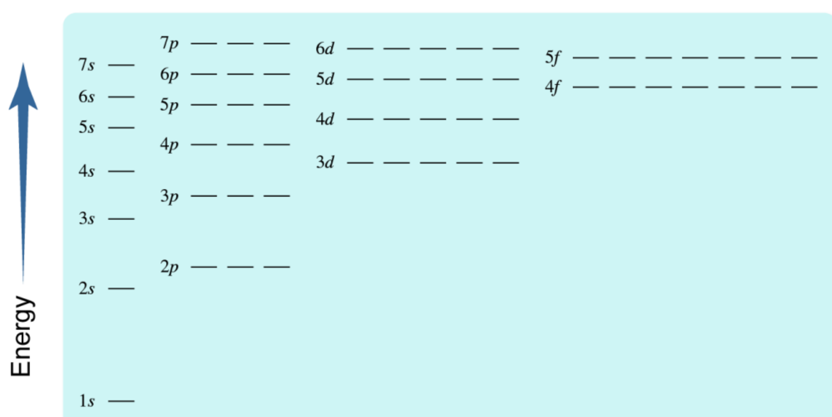
8.C.1. In the energy level diagram below, label or highlight the following:

A. The *n* = 3 shell — This includes all orbitals with *n* = 3.

B. A subshell that can hold 10 electrons — Any of the *d* subshells.

C. An orbital that can hold 2 electrons — Any horizontal line below.

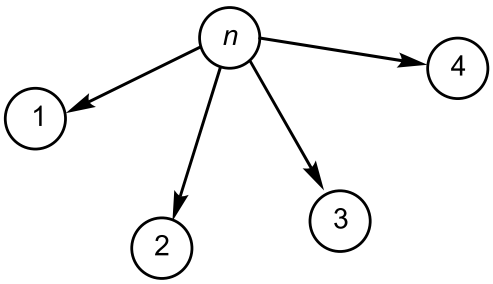
D. Two or more degenerate orbitals — Degenerate orbitals have equal energies.



8.C.2. Applying the Aufbau principle and Hund's rule, number the first ten orbitals to be filled with electrons based on the diagram above.

Start from the lowest-energy subshell (1*s*) and work your way up!

8.C.3. One representation of the rules for quantum numbers involves conceiving of them as a kind of tree, with *n* as the "trunk" and the more specific quantum numbers as "branches." Starting from a representation like the one below, depict the shells, subshells, and orbitals as a tree, labeling each node with its associated quantum number (use letters for ℓ).



8.C.4. Give the name and number of orbitals for each subshell with the following quantum numbers.

A. $n = 3, \ell = 2$ 3*d*, 5 orbitals

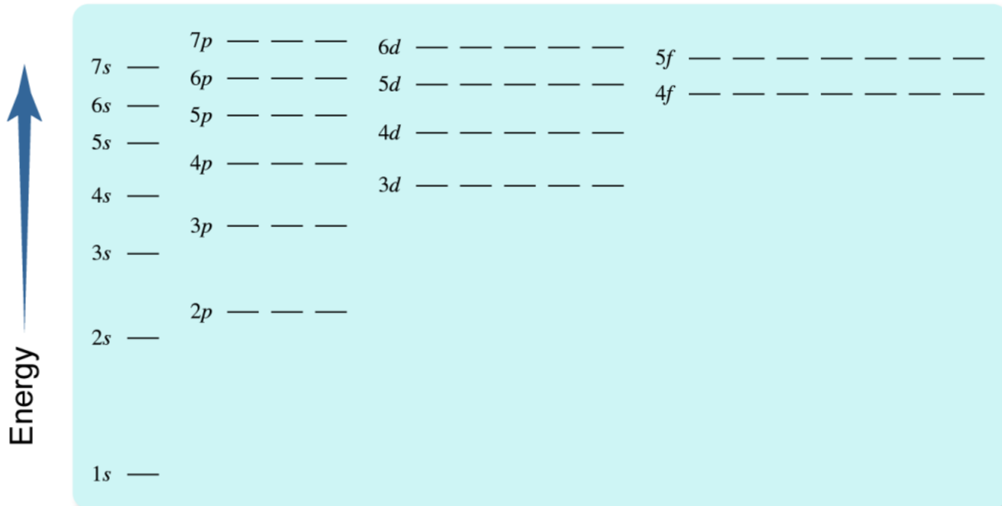
B. $n = 2, \ell = 0$ 2*s*, 1 orbital

C. $n = 4, \ell = 1$ 4*p*, 3 orbitals

D. $n = 5, \ell = 3$ 5*f*, 7 orbitals

Quantum Model of the Atom IV

8.D.1. For the 1*s*, 2*p*, and 3*d* orbitals on the diagram below, indicate the values of *n* and ℓ for each subshell and list the possible values of m_ℓ under each orbital.



8.D.2. Draw the shapes of the 1*s*, 2*p*, and 3*d* orbitals in the table below.

1*s*				
2*p*	2*p*	2*p*		
3*d*	3*d*	3*d*	3*d*	3*d*

Quantum Model of the Atom V

8.E.2. Apply the diagram above to write electron configurations for the following atoms or ions.

A. Lithium atom $1s^22s^1$ = [He]$2s^1$

B. Oxygen atom $1s^22s^22p^4$ = [He]$2s^22p^4$

C. Chloride ion (Cl^-) $1s^22s^22p^63s^23p^6$ = [Ar]

D. Sulfide ion (S^{2-}) $1s^22s^22p^63s^23p^6$ = [Ar]

E. Calcium ion (Ca^{2+}) $1s^22s^22p^63s^23p^6$ = [Ar]

8.E.3. Do you notice anything interesting about the electron configurations of Cl^-, S^{2-}, and Ca^{2+}?

They're all the same (the ions are **isoelectronic**) and they're all equivalent to the configuration of argon, the closest noble gas on the periodic table.

8.E.4. Compare the electron configurations of neutral carbon, silicon, and germanium atoms, all of which are in group 14. What do you notice about their electron configurations?

All are [X]ns^2np^2, meaning all group 14 elements have the same number of valence electrons. The types of compounds they form and their covalent bonding patterns are similar as a result.

Periodicity and Ionic Bonding I

9.A.1. The concept of effective nuclear charge (Z_{eff}) explains a great deal of the periodic variation in atomic properties we encountered in chapter 9. Reasonably accurate values of Z_{eff} can be calculated using Slater's rules. In the equations below, Z is the total nuclear charge, N_{core} and N_{val} are the numbers of core and valence electrons, σ_{core} and σ_{val} are the screening constants of core and valence electrons, and S is the shielding constant. S is a measure of the extent to which other electrons shield a valence electron from the positive charge of the nucleus.

$$Z_{eff} = Z - S$$

$$S = \sigma_{core}N_{core} + \sigma_{val}(N_{val} - 1)$$

$$\sigma_{core} = 0.85,\ \sigma_{val} = 0.35$$

Using a simple picture of the nucleus, core electrons, and valence electrons,
explain why $\sigma_{core} < \sigma_{val}$.

Core electrons are closer to the nucleus than valence electrons and thus directly "interfere" with the "view" that the valence electrons have of the nucleus. Because valence electrons are roughly the same distance from the nucleus, repulsion between them doesn't affect the nuclear charge felt by a valence electron very much.

9.A.2. Choose an element in the table at this link (make sure to avoid elements that are grayed out). Determine the numbers of core and valence electrons in an atom of the element and calculate the effective nuclear charge for a valence electron. Fill these values into the table and look for patterns in Z_{eff}.

See this spreadsheet for the results.

9.A.3. Empirical atomic radii based on the work of Slater are listed here. Add the atomic radius of your element in picometers to the table (if your element has "no data," leave the atomic radius cell blank).

A. Examine the graph of atomic radius and Z_{eff} that appears. Within a period, how does atomic radius depend on Z_{eff}?

   As Z_{eff} increases, atomic radius decreases. Greater effective nuclear charge causes all electrons to be "pulled in" closer to the nucleus.

B. Compare atomic radii for your element and others in its group. The trend moving down a group appears to be opposite the trend observed in A moving across a period. Why?

   Down a group, the valence shell gets bigger (n = 1, 2, 3, …).

9.A.4. All of the following ions are isoelectronic—they have the same electron configuration. Calculate Z_{eff} for a valence electron in each ion and then order them from smallest to largest ionic radius. Explain your ordering.

Se^{2-}, Br^-, Rb^+, Sr^{2+}

smallest Sr^{2+}, Rb^+, Br^-, Se^{2-} **largest**

These ions differ only in the number of protons in the nucleus. Without even calculating Z_{eff}, we can understand intuitively that the nuclear charge must be largest for Sr^{2+} and smallest for Se^{2-}.

Periodicity and Ionic Bonding II

9.B.1. Choose an element in the table at this link (make sure to avoid elements that are grayed out). Using the CRC ionization energies listed here, add the *first* ionization energy of your element to the table in units of electron-Volts (eV). If your element has "no data," leave the ionization energy cell blank.

A. Examine the graph of ionization energy and Z_{eff} that appears. Within a period, how does ionization energy depend on Z_{eff}?

   As Z_{eff} increases, ionization energy increases.

B. Compare ionization energies for your element and others in its group. The trend moving down a group appears to be opposite the trend observed in A moving across a period. Why?

Down a group, the valence shell gets bigger (n = 1, 2, 3, …) and the energies of valence electrons increase, meaning they become *easier* to remove.

9.B.2. The table below lists the main-group elements in period 3 in the rows and successive ionization energies (IE1, IE2, IE3, …) in the columns. In the table, place an "X" where the ionization energy is *much* larger than the one before it for each element. Don't consult a list of ionization energies; consider the number of valence electrons in each atom.

	IE1	IE2	IE3	IE4	IE5	IE6	IE7	IE 8
Na		X						
Mg			X					
Al				X				
Si					X			
P						X		
S							X	
Cl								X

9.B.3. Write and compare balanced chemical equations corresponding to ionization energy and electron affinity. Is the process for which electron affinity is the measured energy absorbed or released the exact reverse of ionization energy? Explain why or why not.

Ionization. $A(g) \rightarrow A^{+}(g) + e^{-}$

Electron capture. $A(g) + e^{-} \rightarrow A^{-}(g)$

Electron capture is *NOT* the reverse of ionization! In electron capture, the *neutral* atom is gaining an electron, not the cation (as in the reverse of ionization).

9.B.4. Find the electron affinity of the element you worked with in question 1 here in units of electron-Volts (eV). Add it to the table of atomic data at this link. Note that electron affinities here are reported as positive numbers with the understanding that an atom accepting an electron is generally exothermic.

A. Examine the graph of electron affinity and Z that appears. Within a period, how does electron affinity generally depend on Z?

As Z_{eff} increases, electron affinity becomes more negative (that is, electron capture becomes more exothermic).

B. Find an exception to the trend in the same row as your element. Write the e⁻ configuration of this element and explain why its electron affinity is not consistent with the trend.

Many different answers here, but key exceptions appear in group 2 (the captured electron would need to go in a *p* subshell by itself) and group 15 (the captured electron is forced to pair, with all other electrons unpaired).

9.B.5. Order each series of ionic compounds below from smallest to largest lattice energy. Identify the key structural difference in each series that allows you to make this determination. Confirm your predictions using the lattice energy calculator.

A. KF, KCl, KBr, KI

smallest KI, KBr, KCl, KF **largest**

B. LiCl, NaCl, KCl, RbCl, CsCl

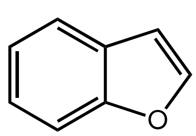
smallest CsCl, RbCl, KCl, NaCl, LiCl **largest**

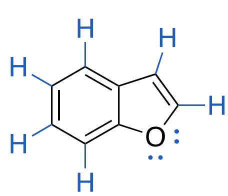
C. MgO, Li_2O, NaCl

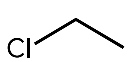
smallest NaCl, Li_2O, MgO **largest**

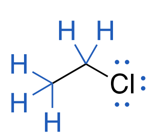
Covalent Bonding I

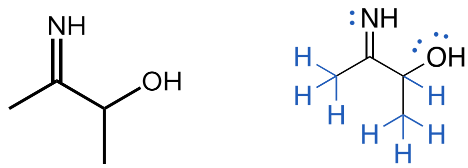
10.A.1. Organic molecules are often depicted using a shorthand notation that omits unshared pairs (lone pairs), carbon atoms, and C–H bonds. Carbon atoms are understood to be located at the intersection points of lines and it is assumed that carbon atoms bear enough bonds to hydrogens to satisfy the octet rule. Lone pairs can be identified by assuming that all atoms in the structure are neutral. Using these ideas, add missing lone pairs and C—H bonds to the structures below.

A. 

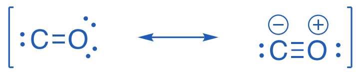


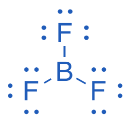
B. 

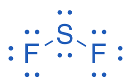


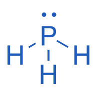
C. 

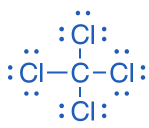
10.A.2. Draw a valid Lewis structure for each of the following molecules.

A. CO 

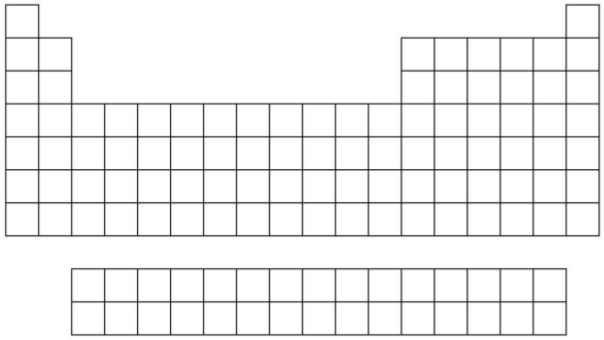
B. BF_3 

C. SF_2 

D. PH_3 

E. CCl_4 

10.A.3. Examine trends in electronegativity here. On the blank periodic table below, indicate how electronegativity changes moving left to right across a period and down a group. Identify the one main-group element that doesn't fit with this trend and explain why it doesn't.

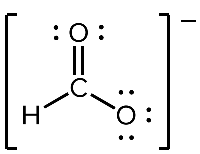


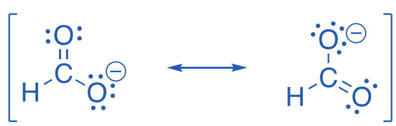
Electronegative increases left to right across a period and decreases going down a group. The lone exception in the main group is hydrogen, but this isn't really an exception if you think of hydrogen as being in group 17…

Covalent Bonding II

10.B.1. The formate ion HCO_2^- has two important resonance structures.

A. One of these structures is drawn below. First, add formal negative charge to the appropriate atom in the structure below. Then, draw the other resonance form, which also includes one C=O bond and one C–O bond.

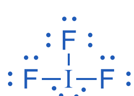




B. Considering electronegativity trends, is the negative charge in formate more likely to reside on carbon or oxygen? Explain.

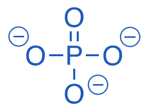
10.B.2. Draw Lewis structures for the following molecules and explain how they violate the octet rule. Include any formal charges and alternative resonance forms.

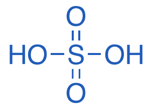
A. IF_3



B. BeH_2

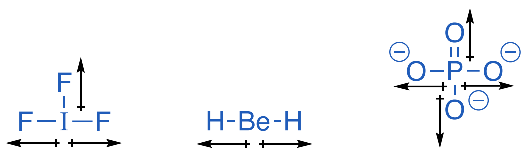
H-Be-H

C. PO_4^{3-} 

D. H_2SO_4 

Covalent Bonding III

10.C.1. For molecules in A – C in the last problem, indicate the more electronegative atom in each bond and use partial charge indicators (δ+ and δ–) to show the orientation of bond dipoles.



Lone pairs have been omitted for clarity.

10.C.2. Thiosulfate ($S_2O_3^{2-}$) reacts with cyanide (CN^-) to form thiocyanate (SCN^-) and sulfite (SO_3^{2-}). A balanced chemical equation for this process is shown below.

$S_2O_3^{2-}(aq) + CN^-(aq) \rightarrow SO_3^{2-}(aq) + SCN^-(aq)$

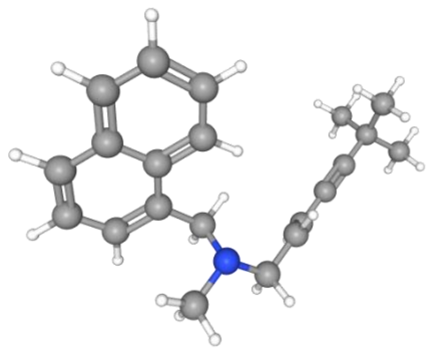
A. Draw Lewis structures for the reactant and product molecules and determine the bonds made and broken in this reaction.

B. Use the bonds made and broken along with bond enthalpies to calculate the enthalpy change of this reaction.

Although this problem is good practice for drawing Lewis structures, resonance complicates the use of bond enthalpies to calculate the reaction enthalpy here. Just draw the Lewis structures in part A, and try to understand how resonance complicates the situation here.

Molecular Shape and Bonding Theories I

11.A.1. Examine the three-dimensional structure of the antifungal terbinafine here. Identify atoms in the structure with linear, trigonal planar, and tetrahedral geometry. Identify an atom with tetrahedral electronic geometry but pyramidal molecular geometry.



The nitrogen atom has tetrahedral electron geometry but pyramidal molecular geometry. Most carbons in the "western half" are trigonal planar; the two carbons involved in the triple bond have linear geometry. The "eastern half" also includes a carbon–carbon double bond, the carbons of which are trigonal planar. Remaining carbons are tetrahedral.

11.A.2. Determine the electronic and molecular geometries of the central atoms in each of the following ions or molecules. Draw Lewis structures for each, determine the number of electron domains around the central atom, write the electronic geometry, and place unshared pairs to determine the molecular geometry.

		electron geometry	molecular geometry
A.	CS_2	linear	linear
B.	SiH_2	trigonal planar	bent
C.	NF_3	tetrahedral	pyramidal
D.	SiF_5^-	trigonal bipyramidal	trigonal bipyramidal
E.	SF_6	octahedral	octahedral

11.A.3. So-called "hypervalent iodine" compounds contain iodine linked to three or more atoms or groups. IF_3 is one of the simplest examples of this kind of compound. What are the electronic and molecular geometries of IF_3?

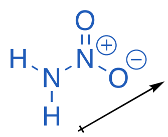
Respectively, trigonal bipyramidal and T-shaped.

11.A.4. Sulfur tetrafluoride (SF_4) is an important fluorinating reagent. Draw a Lewis structure for this compound and determine its electronic and molecular geometries at sulfur.

Respectively, trigonal bipyramidal and see-saw.

Molecular Shape and Bonding Theories II + III

11.B + C.1. The compound nitramide has the formula H_2NNO_2. Draw a Lewis structure for this compound and determine whether it is polar or nonpolar. If the molecule has a permanent dipole moment, draw its direction.



Lone pairs have been omitted for clarity.

11.B + C.2. The following molecules all contain polar covalent bonds. Which are polar molecules and which have no permanent dipole? (a) CCl_4; (b) $CHCl_3$; (c) CO_2; (d) H_2S; (e) SO_2.

The polar molecules are b, d, and e.

Molecular Shape and Bonding Theories IV

11.D.1. Determine the hybridization at the central atom in each of the molecules below.

A. CO_2 *sp*

B. H_2CO sp^2

C. NH_3 sp^3

D. $CHCl_3$ sp^3

11.D.2. Hybrid atomic orbitals are mathematical constructions created by scaling and adding the hydrogenic atomic orbitals (2*s*, 2*p*, etc.) on a single atom. Overlap of hybrid orbitals generates localized bonding orbitals, where we can imagine the electrons in bonds live.

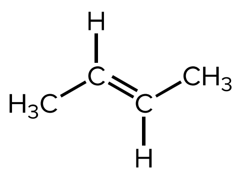
A. A three-dimensional model of allyl alcohol is here. Determine the hybridizations of the carbon atoms in this molecule.

The tetrahedral carbon is sp^3 hybridized and the trigonal planar (doubly bound) carbons are sp^2 hybridized.

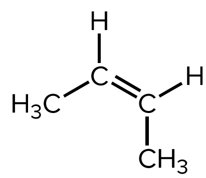
B. Describe the C=C bond in this molecule in terms of valence bond theory. How many sigma and pi bonds are in this bond? What hybrid or atomic orbitals is the sigma bond composed of?

The double bond consists of one sigma bond derived from the overlap of two sp^2 hybrid AOs (one from each carbon) and one pi bond derived from the overlap of two 2*p* AOs (one from each carbon).

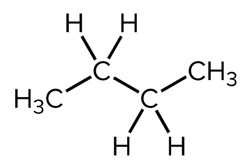
11.D.3. There are two forms of the compound 2-butene that cannot be interconverted except under harsh conditions: *cis*- and *trans*-2-butene. Analogous forms of the related compound butane interconvert extremely rapidly. Use valence bond theory to explain the difference in behavior of butene and butane.



trans-2-butene



cis-2-butene



butane

Rotation around double bonds is slow because such rotation causes the 2*p* orbitals involved in the pi bond to rotate out of alignment, destroying orbital overlap and essentially breaking the pi bond. In butane, which lacks the C=C double bond, rotation around the central C—C single bond (a pure sigma bond) can occur without any change in the overlap of the hybrid AOs that constitute this bond.

Molecular Shape and Bonding Theories V

11.E.1. Draw general shapes for σ, σ*, π, and π* molecular orbitals and list them from lowest to highest energy in general.

11.E.2. Write the molecular electron configuration for ground-state dihydrogen, H_2. What happens to the strength of the H—H bond when H_2 is reduced to H_2^-? Support your argument by calculating the bond orders in H_2 and H_2^-.

$[H_2] = \sigma_{1s}^2$ and $[H_2] = \sigma_{1s}^2\sigma^*_{1s}{}^1$. Reduction of H_2 to H_2^- *weakens* the H—H bond because the newly added electron must occupy an antibonding orbital. The bond order goes from 1 to 0.5 upon reduction.

Molecular Shape and Bonding Theories VI

11.F.1. Write molecular electron configurations for N_2 and O_2 and use them to calculate the bond order in each molecule. Ensure your calculated bond orders align with the Lewis structures of these molecules.

$[N_2] = \sigma_{2s}^2\sigma^*_{2s}{}^2\pi^4\sigma_{2p}^2$ and $[O_2] = \sigma_{2s}^2\sigma^*_{2s}{}^2\sigma_{2p}^2\pi^4\pi^{*2}$. The bond order in N_2 is 3 and the bond order in O_2 is 2; these values agree with the classical Lewis structures for these molecules.

11.F.2. The phenomenon of *s-p* mixing causes a difference in the energetic ordering of molecular orbitals in N_2 and O_2. In your own words, describe *s-p* mixing. What two molecular orbitals are involved and how does *s-p* mixing affect their energies? Why are O_2 and F_2 unaffected by *s-p* mixing?

s-p Mixing involves an interaction between the σ^*_{2s} and σ_{2p} orbitals. These can "mix" (i.e., form two linear combinations, one higher in energy and one lower in energy than the original MOs) and such mixing causes the σ_{2p} orbital to rise in energy. This effect is particularly pronounced when the two orbitals involved are close in energy, which happens when the elements have relatively low electronegativity (N_2 and to the left on the periodic table). As oxygen and fluorine are quite electronegative, the σ_{2p} and σ^*_{2s} orbitals are quite spaced out in energy and as a result, O_2 and F_2 are unaffected by *s-p* mixing.

Molecular Shape and Bonding Theories VII

11.G.1. Using MO diagrams, predict the bond order for the stronger bond in each pair and determine whether each molecule is paramagnetic or diamagnetic. See here.

A. B_2 or B_2^+ — B_2 is paramagnetic and B_2^+ is paramagnetic. The bond in B_2 is stronger.

B. F_2 or F_2^+ — F_2 is diamagnetic but F_2^+ is paramagnetic. The bond in F_2^+ is stronger.

C. O_2 or O_2^{2+} — O_2 is paramagnetic but O_2^{2+} is diamagnetic. The bond in O_2^{2+} is stronger.

D. C_2^+ or C_2^- — C_2^+ and C_2^- are both paramagnetic. The bond in C_2^- is stronger.

Liquids and Solids I

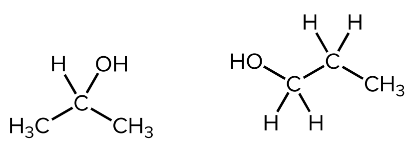
12.A.1. Identify *all* the intermolecular forces (IMFs) in pure samples of each compound below. If more than one force is operating, identify the predominant IMF.

A. CH_3I London forces, dipole-dipole forces (predominant)
B. CH_3CH_2OH London forces, dipole-dipole forces, hydrogen bonding (predominant)
C. Br_2 London forces (predominant)
D. CH_2Cl_2 London forces, dipole-dipole forces (predominant)

12.A.2. Explain why, under ordinary conditions, elemental fluorine and chlorine both exist as gases, bromine exists as a liquid, and iodine exists as a solid.

London forces become stronger as molecular size and surface area increase. The stronger London forces in dibromine and diiodine result in their existing in condensed phases at room temperature.

12.A.3. Suggest a reason why the boiling point of 2-propanol (77 °C) is much lower than the boiling point of 1-propanol (97 °C).



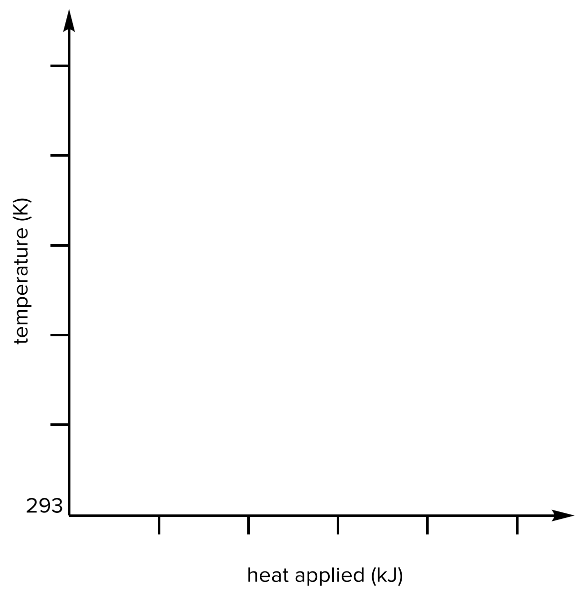
2-propanol 1-propanol

There could be a couple of things going on here. The terminal –OH group in 1-propanol may be more "free" to hydrogen bond than the more crowded internal –OH group in 2-propanol. The "long and skinny" 1-propanol may also display stronger London forces than the spherical 2-propanol.

Liquids and Solids II

Try these on your own!

12.B.1. Use thermochemical data on the NIST Chemistry WebBook to generate a heating curve for one of the following solvents: (a) ethanol; (b) diethyl ether; (c) tetrahydrofuran; (d) dichloromethane. Try to draw the heating curve to scale. Assume all heating happens at a constant pressure of 1 atmosphere and that 1 mole of substance is being heated; extend the curve at least 40 K above the boiling point. For dichloromethane gas, assume c_P = 55 J/mol·K (a good approximation below 350 K).



12.B.2. Create a phase diagram for bromine (Br_2) using the following information. (reference video)

A. The critical point occurs at (102 atm, 315 °C).
B. The triple point occurs at (0.007 atm, –33 °C).
C. The normal boiling point is 59 °C.
D. The density of the solid is greater than the density of the liquid at all pressures.

Liquids and Solids III

12.C.1. Examine the unit cells linked below. For each, determine the number of atoms inside each unit cell (*only* inside the bounding box!) and calculate the packing factor as the percentage of space inside the unit cell filled by atoms. Examples of metals that crystallize in each lattice type are listed; using the identity of the metal and its crystal structure, calculate the density of the metal.

A. Cubic close packing (ccp) or face-centered cubic (fcc) | silver metal (Ag)

The fcc unit cell contains 4 atoms; the volume taken up by these atoms is four times the atomic volume, or $(16/3)\pi r^3$. The volume of the unit cell is $a^3 = (4r/2^{1/2})^3$. The packing factor is thus $(16/3)\pi r^3 / (4r/2^{1/2})^3 = 0.740$ or 74.0%. The density of silver metal based on this packing is 10.6 g/mL.

B. Simple cubic | polonium metal (Po)

The simple cubic unit cell contains 1 atom; the volume taken up by this atom is the atomic volume, or $(4/3)\pi r^3$. The volume of the unit cell is $a^3 = (2r)^3$. The packing factor is thus $(4/3)\pi r^3 / (2r)^3 = 0.524$ or 52.4%. The density of polonium metal based on this packing is 9.15 g/mL.

C. Body-centered cubic (bcc) | sodium metal (Na)

The bcc unit cell contains 2 atoms; the volume taken up by these atoms is two times the atomic volume, or $(8/3)\pi r^3$. The volume of the unit cell is $a^3 = (4r/3^{1/2})^3$. The packing factor is thus $(8/3)\pi r^3 / (4r/3^{1/2})^3 = 0.680$ or 68.0%.

12.C.2. Zinc metal crystallizes in the hexagonal close packing (hcp) structure, which has a packing factor equal to that of fcc. Compare and contrast this structure with the fcc structure. How are the structures different despite their equal packing factors?

Layering in the hcp structure is different—it layers in an ABAB… way, while fcc layers in an ABCABC… way. Not terribly important to know, but this provides another way of looking at the fcc lattice!

Thermodynamics of Solutions

13.A.1. Describe the process of solution formation in terms of interactions between solute and solvent particles. For each step of the process, indicate whether the step is likely to be endothermic or exothermic and whether the change in entropy (ΔS) is likely to be positive or negative.

Solute-solute and solvent-solvent interactions are disrupted (endothermic but positive ΔS) and solute-solvent interactions are created (exothermic but negative ΔS).

13.A.2. Draw molecular-level depictions of the solute particles and their immediate environment when $CaCl_2$ is dissolved in water. Pay careful attention to the orientation of the water molecules!

Try this one on your own! See here for some relevant images.

13.A.3. One approach to thinking about solution thermodynamics for ionic solutes in water involves conceiving of the solution process as ionization followed by hydration:

$MX(s) \rightarrow M^+(g) + X^-(g)$	ionization
$M^+(g) \rightarrow M^+(aq)$	hydration of M^+
$X^-(g) \rightarrow X^-(aq)$	hydration of X^-
$MX(s) \rightarrow M^+(aq) + X^-(aq)$	solution formation

Determine the enthalpy of solution for MgF_2 using the following data. Is the formation of an aqueous solution of magnesium fluoride endothermic or exothermic?

Lattice energy of MgF_2: 2908 kJ/mol
Enthalpy of hydration of Mg^{2+}: –1926 kJ/mol
Enthalpy of hydration of F^-: –524 kJ/mol

+2908 kJ – 1926 kJ + 2(–524 kJ) = –66 kJ. This dissolution process is expected to be slightly exothermic. Note that the enthalpy of hydration of F^- was multiplied by 2 because 2 fluoride ions dissolve per formula unit of MgF_2.

13.A.4. Indicate which molecule in each pair is more likely to dissolve in water, drawing Lewis structures where necessary. Explain differences in solubility using intermolecular forces.

A. **$NaNO_3$** or C_6H_6

Sodium nitrate is an ionic compound; each ion can engage in ion-dipole interactions with water molecules in solution. C_6H_6 is nonpolar and cannot engage in strong IMFs with water as a result.

B. C_2H_6 or **CH_3CH_2OH**

Ethanol can hydrogen bond with water but C_2H_6 is nonpolar.

C. C_5H_{12} or **$CH_3C(=O)CH_3$**

Acetone can engage in hydrogen bonding with water (via its oxygen, a hydrogen-bond acceptor) but C_5H_{12} is nonpolar.